

From Conflict to Human Capital?

Evidence from Colombia's Peace Process

David Bernal
University of Houston

Draft: August 2026

Abstract

Does peace move students through the educational pipeline, or does it only remove one of the constraints they face? I link administrative records for 4.90 million regular, second-semester Saber 11 examinees from 2010 to 2018 to first-enrollment records from Colombia's National Higher Education Information System (SNIES), university-level Saber Pro, and harmonized technical and technological examination records through 2024. I combine these student outcomes with a municipality-year conflict panel and compare places with historical FARC presence with historical ELN municipalities, as well as PDET territories with non-PDET municipalities. Conflict falls sharply in historical FARC areas after negotiations begin: the pooled core-conflict index declines by 0.42 standardized-index units relative to ELN areas, with stable pre-trends. The corresponding educational changes are far less general. The negotiation-period designs show no increase in higher-education entry or overall exam-stage progression. After the 2016 agreement, PDET-origin students become 0.96 percentage points more likely to reach university-level Saber Pro within six years, but the effect is absent in the FARC-ELN comparison and cannot be separated from the nonrandom selection and later policy treatment of PDET territories. SNIES enrollment, short-cycle progression, Saber 11 achievement, value added, timing, financing, migration, and field choices do not show a common improvement across designs. The evidence therefore rejects a simple transmission mechanism from security to human capital. Peace substantially reduced organized violence, while the limited downstream gains are consistent with a need for complementary capacity and appear, at most, concentrated in territories prioritized after the agreement.

Keywords: peace agreements, armed conflict, education, higher education, Colombia, human capital
JEL codes: D74, I21, I23, O15

1 Introduction

The end of a civil conflict can alter educational investment through several channels. Lower violence reduces the danger of traveling to school, weakens recruitment by armed groups, limits displacement, and lengthens the horizon over which families value schooling. Peace may also improve market access and encourage institutions to enter previously isolated places. Yet disarmament does not itself provide teachers, universities, roads, credit, or effective local government. Security can relax an access constraint while leaving the productive capacity of the education system unchanged.

Colombia offers an unusually informative setting in which to separate these margins. Formal negotiations between the national government and the Revolutionary Armed Forces of Colombia, the FARC, began in 2012. The FARC declared an indefinite unilateral ceasefire in December 2014, and the final agreement was signed in November 2016. These events separated negotiation, de-escalation, and implementation. They also generated meaningful spatial variation. Some municipalities had long histories of FARC presence, while others had historical presence of the National Liberation Army, the ELN, which did not demobilize under the FARC agreement. Historical presence is defined from actor-attributed municipal events in the Conflict and Violence Panel assembled by the Centro de Estudios sobre Desarrollo Económico (CEDE), the economics research center at Universidad de los Andes (CEDE, 2022). A municipality must record activity in at least eight of the thirteen years from 1996 through 2008. The FARC and ELN comparison is mutually exclusive: a FARC municipality meets this threshold only for the FARC, and an ELN municipality meets it only for the ELN. Municipalities meeting the threshold for both groups or for neither group are omitted. The data, classification, and sample restrictions are detailed in [Appendix Section D.1.2](#). In 2017, the government additionally designated 170 conflict-affected, poor, and institutionally weak municipalities for the Programas de Desarrollo con Enfoque Territorial, or PDET (Colombia, 2017; United Nations, 2017).

This paper asks how the peace process changed the path from secondary-school exit to higher education. The analysis begins with students taking Saber 11, Colombia's secondary exit examination, in the regular second-semester session. It then follows them to first enrollment through a crosswalk linking Saber 11 to the *Sistema Nacional de Información de la Educación Superior* (SNIES), Colombia's National Higher Education Information System.¹ Later progression is measured with two national exit-stage examinations administered by ICFES: Saber TyT for short-cycle technical and technological programs, and Saber Pro for professional university programs. These examinations are normally taken after completing at least 75 percent of program credits. The linked records therefore distinguish first entry from reaching the final portion of a short-cycle or university program.

This distinction addresses an important measurement problem. Saber Pro is taken by students who have completed at least 75 percent of a professional degree, and it is normally required for graduation. Appearance in Saber Pro is therefore a strong measure of university near-completion, but it is neither first enrollment nor formal graduation. Technical and technological students were included in files distributed with Saber Pro before ICFES separated Saber TyT reporting in 2016. Counting all records from the pre-2016 files and only Saber Pro afterward would generate a mechanical break. I harmonize the series by combining technical and technological Saber Pro records before the split with Saber TyT afterward, while keeping professional university Saber Pro separate in every cohort.

The empirical analysis uses two spatial comparisons and two policy windows. The historical FARC

¹SNIES is the Ministry of National Education's administrative information system for higher-education institutions, programs, enrollment, and other sector records. The public Saber 11-SNIES crosswalk used here identifies each matched student's first observed higher-education institution, program, and enrollment semester.

versus ELN design compares conflict-exposed municipalities that differed in whether the dominant insurgent organization signed the agreement. It is the most direct design for the security treatment but has only 41 ELN comparison municipalities. The PDET comparison has substantially greater power and captures territories at the center of implementation. It is not a quasi-random treatment: PDET status was assigned after the agreement using conflict, poverty, rurality, institutional weakness, and illicit-economy criteria. I therefore interpret PDET estimates as the relative trajectory of prioritized places and not as the effect of designation before 2017.

The 2012 window asks whether the onset of negotiations changed educational decisions. The 2016 window asks whether cohorts making postsecondary transitions at the time of the final agreement experienced different trajectories. The second window has a stronger trend diagnostic because the public Saber 11-SNIES crosswalk begins with the 2010 Saber 11 cohort. A 2012 event study therefore has only one pre-treatment coefficient, while a 2016 event study has five. The paper reports both rather than treating a statistically insignificant one-lead pre-trend test as strong validation.

Four findings organize the paper. First, the security treatment was real. Historical FARC municipalities experience a pooled decline of 0.42 standardized-index units in the core conflict index relative to historical ELN municipalities after negotiations begin. The corresponding log event-rate estimate is -0.66 . The pre-treatment coefficients are jointly insignificant, and the decline persists through 2018. The educational results cannot therefore be dismissed as consequences of a peace process that failed to reduce the measured forms of insurgent violence.

Second, the negotiation-period comparisons show no broad postsecondary dividend. In the common 2010 to 2018 sample, neither design shows a statistically significant increase in SNIES enrollment within one, two, or six years. Neither design shows a significant increase in overall exam-stage progression, defined as reaching Saber TyT or Saber Pro within six years. PDET short-cycle progression instead falls by 1.49 percentage points, while professional university progression is statistically unchanged. The FARC versus ELN estimates provide no statistically significant evidence of greater short-cycle, university, or overall progression.

Third, the post-agreement PDET estimates are more favorable at the professional university margin. Relative to the 2015 cohort, PDET-origin students are increasingly likely to reach university-level Saber Pro in 2016, 2017, and 2018. The pooled estimate is 0.96 percentage points with a standard error of 0.43, and a joint test does not reject the 2010 to 2014 pre-treatment coefficients. However, the analogous FARC versus ELN estimate is 0.44 points with a standard error of 0.58 and rejects parallel pre-trends. SNIES entry does not increase consistently, and the PDET estimate for any tertiary near-completion has differential pre-trends. The university result is therefore suggestive evidence of a relative gain in territories later prioritized by the state, not a clean estimate of the security effect of the peace agreement.

Fourth, the remaining outcomes do not reveal a broad human-capital transformation. PDET municipalities have more Saber 11 test taking after the agreement but lower mean mathematics and verbal scores, a pattern consistent with changing selection into the examination. The FARC versus ELN score estimates remain near zero. Conditional time to Saber Pro, value added between Saber 11 and Saber Pro, financing, migration, institution type, and field of study do not improve consistently across both designs. These selected-completer outcomes are informative about mechanisms, but they cannot overturn the full-cohort evidence.

The paper contributes to three literatures. First, it extends research on the human-capital consequences of conflict from school attendance and achievement to multiple stages of the postsecondary pipeline. Previous Colombian research studies school enrollment, child labor, Saber 11 achievement, or school-level responses to de-escalation (Rodríguez and Sánchez, 2012; Gómez Soler, 2016, 2017;

Namen et al., 2021; Montaña-Bardales and Palacios, 2025). In contrast, this paper follows individual recent school leavers into first higher-education enrollment and then toward the exit stage of both short-cycle technical or technological programs and professional university programs. Second, it verifies that the peace process changed measured insurgent violence before asking whether educational outcomes changed along the same spatial contrast. This separates an absent educational response from an absent security shock. Third, the paper makes the pre- and post-2016 examination records comparable by harmonizing the reporting split between Saber Pro and Saber TyT (ICFES, 2016). It also keeps the full eligible Saber 11 cohort in every denominator: nonmatches and students who do not reach the relevant exam stage within the window are coded as zeros rather than dropped.

The central conclusion is deliberately narrower than the claim that peace had no educational effect. Peace clearly reduced conflict, and municipalities later selected for PDET show a suggestive post-agreement rise in professional university near-completion. Because that rise begins before formal PDET designation, it is not an estimated effect of the program. What the data do not show is a general security-driven improvement in enrollment, short-cycle progression, university progression, achievement, speed, or value added. Security is an input into human-capital production. It is not a substitute for the institutions and resources that convert safer lives into sustained educational attainment.

2 Related Literature

This paper first relates to the large literature on the human-capital consequences of armed conflict. Conflict can reduce educational investment through displacement, recruitment, child labor, trauma, household income losses, school closures, and the destruction or withdrawal of public services (Blattman and Miguel, 2010; Justino, 2012, 2016). Microeconomic studies document persistent schooling losses among conflict-exposed cohorts in Tajikistan, Guatemala, and Peru (Shemyakina, 2011; Chamarbagwala and Morán, 2011; León, 2012). This evidence establishes that insecurity can damage both the household demand for education and the institutional capacity to supply it. It does not, however, establish that removing violence reverses those losses automatically or symmetrically.

A second literature examines education during Colombia's armed conflict. Exposure to violence has been associated with lower school enrollment, greater child labor, and weaker achievement on the Saber 11 secondary exit examination (Rodríguez and Sánchez, 2012; Gómez Soler, 2016, 2017). More recent work studies de-escalation itself. The FARC ceasefire changed school attendance, dropout, achievement, and the composition of students reaching the examination margin (Namen et al., 2021; Avendaño Santiago, 2021; Montaña-Bardales and Palacios, 2025). These studies focus primarily on children still in primary or secondary school. The present paper examines a later transition that requires additional household resources and institutional inputs: whether recent secondary-school leavers enter higher education and progress toward the end of a postsecondary program.

Third, the paper contributes to research on the broader consequences of Colombia's peace process. The agreement sharply reduced FARC-related violence, but its local economic effects were heterogeneous. Entrepreneurship initially increased, broad economic activity remained limited where effective state entry was weak, and the end of conflict also generated unintended environmental responses (Bernal et al., 2024; Fajardo-Steinhäuser, 2024; Prem et al., 2020). This literature makes clear that a security dividend need not translate one-for-one into development outcomes. By measuring conflict, first enrollment, short-cycle progression, professional university progression, and achievement in the same study, this paper identifies the stages at which transmission toward human capital does or does not appear.

Finally, the analysis speaks to the literature on higher-education access and persistence. Postsec-

ondary investment is sensitive to income risk, uncertainty about costs and academic fit, and the probability of completing a credential (Kazianga, 2012; Stange, 2012; Athreya and Eberly, 2021). In Colombia, expanding institutional supply, student credit, and merit-based aid can alter enrollment and progression, particularly for low-income students (Carranza and Ferreyra, 2019; Melguizo et al., 2016; Londoño-Vélez et al., 2020). These findings emphasize that safer conditions are only one input into postsecondary attainment. Preparation, available places, distance, finance, information, and institutional support may remain binding after violence declines. The contribution here is to connect these higher-education margins to a major peace episode using linked individual administrative records rather than inferring postsecondary access from school outcomes alone.

3 Institutional Background

3.1 Negotiation, ceasefire, and agreement

The FARC emerged in the 1960s and became Colombia's largest guerrilla organization. Its presence was concentrated in rural areas where roads, legal markets, public services, and state authority were often weak. Armed conflict affected education through violence, displacement, recruitment, teacher turnover, school closure, and illicit labor-market opportunities.

Public negotiations began in 2012, but negotiation did not immediately end fighting. The FARC initiated an indefinite unilateral ceasefire on December 20, 2014. A bilateral ceasefire followed in 2016, and the revised Final Agreement was signed on November 24 and entered into force on December 1, 2016 (United Nations, 2016, 2017). These dates create different educational exposures. A 2012 Saber 11 cohort made postsecondary choices during uncertain negotiations. A 2016 cohort entered its postsecondary years as the agreement took effect.

The agreement combined disarmament and political incorporation with rural reform, crop substitution, victim reparations, and territorial development. Implementation nevertheless began in places with limited fiscal resources and weak public-service networks. Other armed organizations could also enter territories vacated by the FARC. The agreement removed a central source of violence without instantly producing complete state presence.

3.2 PDET territories

Decree Law 893 of 2017 created the PDET as the territorial vehicle for implementing rural reform. It selected 170 municipalities organized into 16 subregions. Selection reflected armed conflict, poverty, institutional weakness, rurality, and illicit economies (Colombia, 2017). These municipalities are substantively important because they concentrate the population for whom a peace dividend was most needed. They are not a randomly assigned treatment group.

This timing is essential. A PDET event study around 2012 describes the trajectory of places later selected for the program. A PDET comparison around 2016 combines the agreement, subsequent expectations, and the early phase of targeted territorial policy. I do not call either specification the causal effect of PDET designation before 2017. The FARC versus ELN comparison more directly targets the change in security, conditional on ELN municipalities providing a valid counterfactual.

4 Data and Measurement

4.1 Saber 11 cohorts

The starting population comes from administrative examination records of the Colombian Institute for the Evaluation of Education, ICFES. Saber 11 is the national examination taken near the end of secondary school. I retain regular students who take the examination in the second semester of each year. This is the standard session for students finishing grade 11 in Colombia's traditional academic calendar. The restriction makes the starting population a more homogeneous group of recent school leavers rather than a mixture of recent graduates, adult examinees, and off-cycle test takers.

Municipality is the student's residence when taking Saber 11. The analysis therefore asks whether youth originating in a municipality enter and progress through higher education, regardless of whether they later migrate to study. Municipality codes are harmonized over time, and observations without a usable origin code are excluded before estimation.

The main postsecondary-pipeline sample contains 4,904,919 students from the 2010 to 2018 Saber 11 cohorts. All figures and tables comparing SNIES entry, Saber TyT, and Saber Pro use this common sample because the public Saber 11-SNIES crosswalk begins with the 2010 cohort. A larger sample of 5,354,444 students appears only in the explicitly labeled university-only robustness analysis. That sample adds the 2009 cohort because the Saber Pro linkage, unlike the SNIES crosswalk, covers 2009. I never code 2009 students as non-enrollees: their enrollment status is unobserved in the released SNIES crosswalk.

4.2 SNIES first enrollment

The Saber 11-SNIES crosswalk identifies a student's first observed higher-education institution, program, and semester. Of the 4.90 million students in the common sample, 2,849,218 match to SNIES. I define entry within one year as first enrollment during the first two semesters after Saber 11, entry within two years as the first four semesters, and entry within six years as the first twelve semesters. The enrollment semester must be strictly later than the student's Saber 11 semester. For 44,972 linked students, the first reported enrollment occurs in or before the Saber 11 semester. These records are not necessarily erroneous: they may reflect earlier higher-education enrollment, delayed Saber 11 testing, or semester-level timing inconsistencies. They remain in the eligible cohort but do not count as post-Saber 11 entry because they cannot identify a transition occurring after the baseline examination.

SNIES provides a direct measure of higher-education access. It separates a decision to enter higher education from the much later event of appearing in an exit-stage examination. The released crosswalk used here does not provide a complete semester-by-semester history of enrollment interruptions or formal graduation.

4.3 Saber Pro, Saber TyT, and harmonization

Saber Pro is designed for students in professional university programs who have completed at least 75 percent of their credits. Taking the examination is normally a graduation requirement ([ICFES, 2026](#)). I therefore describe appearance as university near-completion or university exam-stage progression, not graduation. The main indicator equals one when a student appears in university-level Saber Pro within six years of Saber 11. Students never observed in Saber Pro, observed only outside the six-year window,

or observed only in a short-cycle program receive zero.

The six-year window balances substantive and statistical considerations. Professional programs normally require at least four years, and many require five. Saber Pro eligibility before formal graduation makes some appearances earlier than program completion, while interruptions and delayed entry make others later. Six years therefore captures standard duration plus moderate delay. It also gives every cohort through 2018 exactly the same maximum follow-up through 2024. An unrestricted window would mechanically favor older cohorts, while a four-year window would classify many ordinary professional trajectories as failures. The complementary university-only analysis also reports an outcome restricted to years four through six; it differs from the within-six-year definition for only 7,327 students and reaches the same qualitative conclusion.

Technical and technological programs require a separate construction. These shorter vocational and applied programs are roughly analogous to degrees offered by U.S. community colleges, although Colombia's institutional structure and credentials differ. Before the 2016 reporting split, these students appear in files historically distributed with Saber Pro. From 2016 onward, ICFES reports them through Saber TyT (ICFES, 2016). I identify technical and technological program levels in the historical files and combine them with linked Saber TyT records after the split. The resulting short-cycle indicator equals one when a student reaches the relevant examination stage within six years. The any-tertiary indicator is the union of university and harmonized short-cycle progression. The common sample contains 818,015 university-level Saber Pro appearances, 671,314 harmonized short-cycle appearances, and 1,434,778 students reaching either exam stage within six years.

All binary outcomes retain the full eligible Saber 11 cohort. This denominator is central. A student who never appears in a linked examination, appears only in an ineligible program type, or appears outside the relevant window is a zero. The analysis does not condition the extensive margin on successful matching to a later examination.

4.4 Conflict, secondary achievement, and conditional outcomes

For the annual conflict outcomes, I organize ten event-level administrative files from the Colombian Ministry of Defense's National Police into a balanced municipality-year panel for 2007 to 2018. Population denominators come from CEDE's municipal General Characteristics Panel. These annual outcome files are distinct from CEDE's Conflict and Violence Panel, which is used only to define historical FARC and ELN presence. The core conflict index uses five event families with complete coverage: subversive actions, attacks on bridges and roads, kidnappings, terrorism, and oil-pipeline bombings. Each count is converted to events per 100,000 residents, transformed using $\log(1 + x)$, standardized over the municipality-year panel, and equally weighted. Homicide and extortion are analyzed separately because they include broader criminal activity.

The secondary-school analysis aggregates Saber 11 records to municipality by year. Outcomes include test takers per thousand residents, represented schools per 100,000 residents, and cohort-standardized mathematics and verbal scores. ICFES redesigned Saber 11 in the second semester of 2014. Language was replaced by critical reading; mathematics remained an examination domain but its structure and score scale also changed (ICFES, 2014). Raw scores are therefore not compared across the redesign. I standardize scores within each national examination cohort, and the pooled post-agreement analysis uses only the redesigned examination period.

Among students observed in university-level Saber Pro, I examine elapsed time, defined as calendar years from the Saber 11 semester to the first observed Saber Pro appearance. I also examine reported

semester, credit and scholarship financing, work, migration, institution type, field, and residual value added. These are conditional composition outcomes. Peace can change who enters the observed near-completion sample, so conditional coefficients need not equal effects for the full eligible Saber 11 cohort.

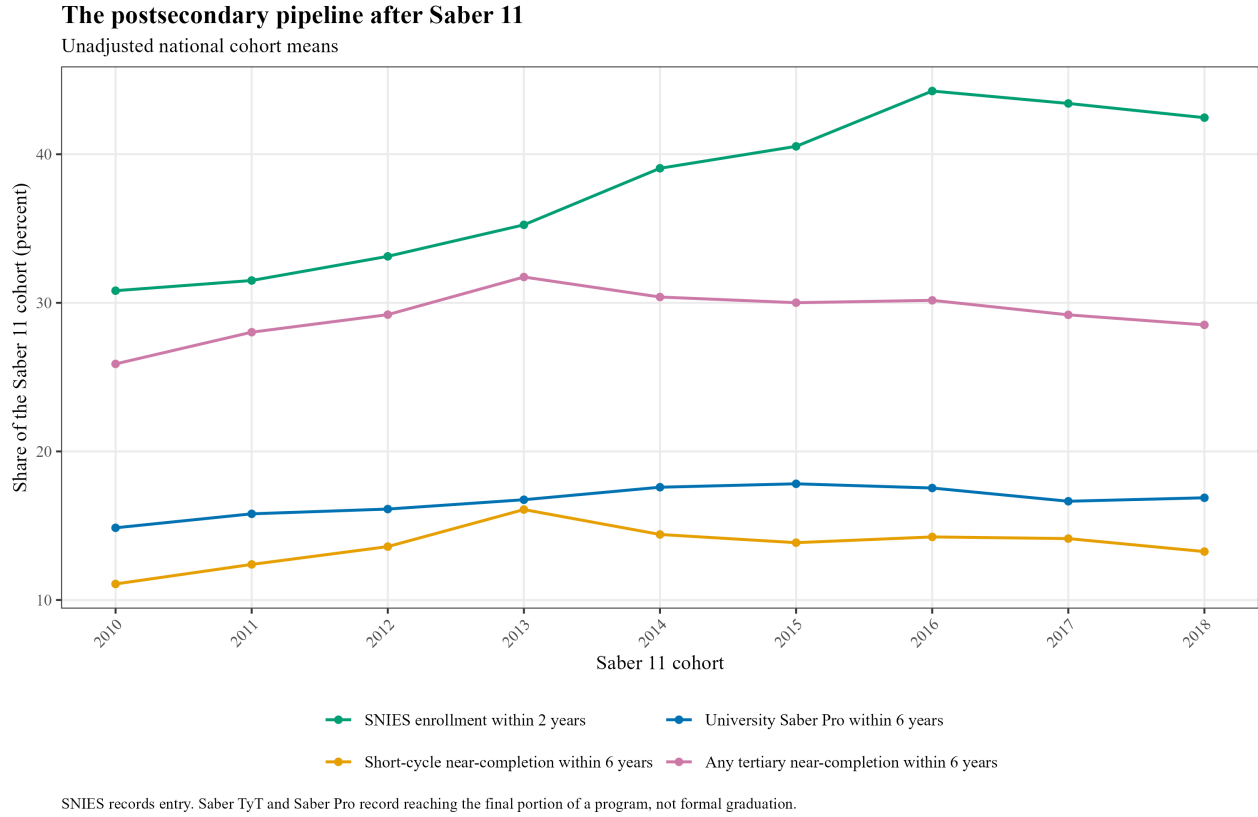


Figure 1: The postsecondary pipeline after Saber 11

Notes: Lines report unadjusted national means for the 2010 to 2018 Saber 11 cohorts. SNIES measures first enrollment after Saber 11. Saber Pro and the harmonized Saber TyT series measure reaching the final portion of a program, not formal graduation. Every outcome uses the full eligible Saber 11 cohort as its denominator.

5 Empirical Strategy

For each student outcome, I estimate the event-study model

$$Y_{imt} = \alpha_m + \lambda_t + \sum_{k \neq b} \beta_k (\mathbb{1}\{t = k\} \times T_m) + X_i' \gamma + \varepsilon_{imt}, \quad (1)$$

where Y_{imt} is an outcome for student i from municipality m in Saber 11 cohort t , α_m are municipality fixed effects, λ_t are cohort fixed effects, and T_m is PDET status or historical FARC presence. X_i contains cohort-standardized Saber 11 mathematics and reading scores in the common pipeline specifications. Standard errors are clustered by municipality. Binary-outcome coefficients are reported in percentage points.

For the negotiation window, the omitted cohort is $b = 2011$ and post-treatment cohorts begin in 2012. For the agreement window, $b = 2015$ and the post period begins in 2016 because these students make their postsecondary transition as the final agreement enters into force. To summarize each event study, I

also estimate the pooled model

$$Y_{imt} = \alpha_m + \lambda_t + \delta(T_m \times Post_t) + X'_i\gamma + \varepsilon_{imt}. \quad (2)$$

Here $Post_t$ is an indicator, not a sum of outcomes. In the common 2010 to 2018 sample, it equals one for cohorts 2012 to 2018 in the negotiation specification and zero for 2010 to 2011. In the agreement specification, it equals one for 2016 to 2018 and zero for 2010 to 2015. Thus, δ compares the average treated-control difference across all post-period cohorts with the average difference across all pre-period cohorts, after removing municipality and cohort fixed effects and adjusting for X_i . The 2012 coefficient summarizes the negotiation and de-escalation period, while the 2016 coefficient compares cohorts transitioning after the agreement with earlier cohorts. Treatment-group status is fixed over time; the two windows test different policy breaks rather than staggered adoption.

Trend diagnostics are reported for every pooled result. The SNIES crosswalk begins in 2010, so the 2012 common-sample event study has only one lead. Failure to reject that single coefficient is weak evidence and is not described as proof of parallel trends. The 2016 event study jointly tests the 2010 to 2014 coefficients relative to 2015. The complementary 2009 to 2018 university-only analysis provides two leads for the 2012 window and serves as a robustness exercise.

The identifying assumption differs by comparison. For FARC versus ELN, the assumption is that, absent the FARC peace process, the educational gap between the two sets of municipalities would have remained stable after common national shocks. For PDET versus non-PDET, the corresponding assumption is that the gap between municipalities later prioritized and all other municipalities would have remained stable. This second assumption is especially demanding because PDET municipalities were selected for conflict, poverty, rurality, institutional weakness, and illicit economies. I therefore regard an estimate as more credible when its pre-treatment coefficients are stable, its timing matches the policy window, and related outcomes or the alternative spatial comparison tell a compatible story.

The security-validation models are estimated at the municipality-year level with municipality and year fixed effects and equal municipality weights. Historical FARC presence requires attributed activity in at least eight of the thirteen years from 1996 through 2008 and no equivalent historical ELN classification. The comparison sample contains 216 mutually exclusive FARC municipalities and 41 ELN municipalities. [Appendix Section D.1.2](#) provides the detailed construction.

6 Results

6.1 Conflict outcomes and validation of the security shock

Figure 2 begins with the outcome closest to treatment; [Appendix Section D.1](#) details the index and its source variables. Historical FARC municipalities experience a large decline in the standardized core conflict index relative to historical ELN municipalities after negotiations begin. The pooled post-2012 coefficient is -0.422 standardized-index units, and the corresponding estimate for the log population-normalized core-event rate is -0.656 . Both are statistically significant at the 1 percent level. Joint tests fail to reject equality of the 2007 to 2010 pre-treatment coefficients, with p values of 0.462 and 0.491. Thus, for these conflict outcomes, I do not detect differential pre-trends between historical FARC and ELN municipalities before negotiations. This supports, but does not prove, the parallel-trends assumption for the security comparison. The relative decline begins in 2012, becomes larger in 2013, and persists through 2018.

Did negotiation change conflict intensity?

Event-study estimates for the standardized core conflict index

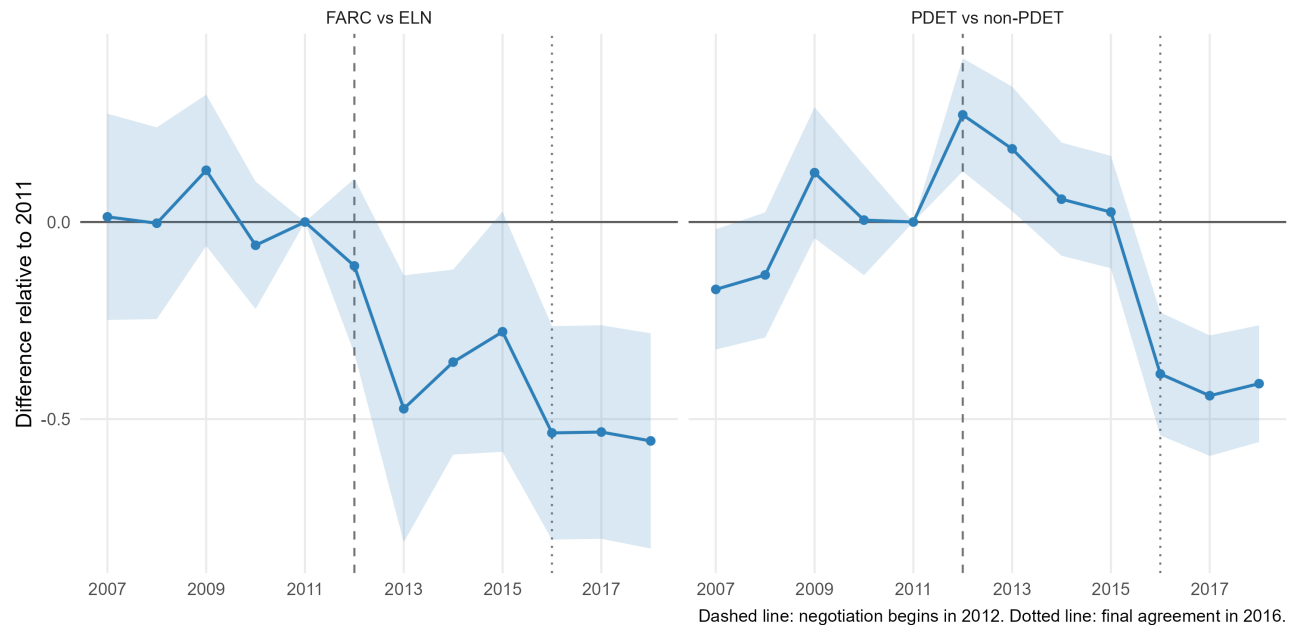


Figure 2: Conflict intensity in historical FARC and ELN municipalities and PDET territories

Notes: The outcome is the standardized core conflict index constructed from subversive actions, attacks on bridges and roads, kidnapping, terrorism, and oil-pipeline bombing. Models include municipality and year fixed effects and omit 2011. Bands are 95 percent confidence intervals from municipality-clustered standard errors.

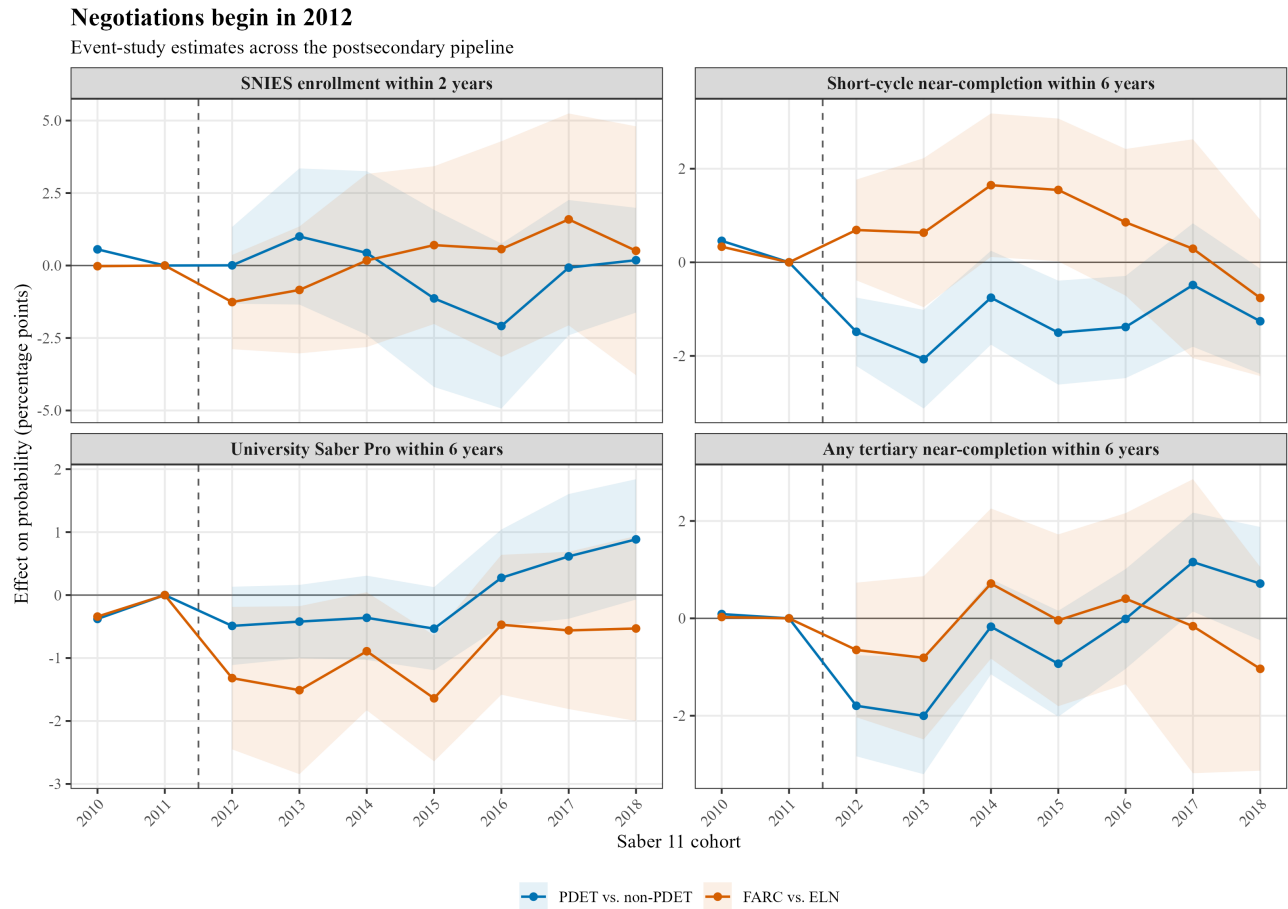
The PDET pattern is different. There is little pooled relative change during negotiations, followed by a large decline after the agreement. The post-2017 conflict-index coefficient is -0.423 and is statistically significant at the 1 percent level. However, the PDET pre-treatment coefficients are also jointly significant. Parallel trends therefore do not hold for this comparison and outcome, so the post-agreement decline is descriptive evidence of de-escalation in prioritized places rather than a clean causal estimate. Broader safety outcomes do not move uniformly: homicide and extortion sometimes diverge from the core guerrilla-related index, consistent with criminal substitution, reporting changes, or entry by other armed groups.

This security validation is a crucial benchmark. The FARC versus ELN estimates establish a large and persistent change in measured insurgent violence with credible pre-treatment dynamics for the conflict outcomes. The analysis can therefore ask whether educational behavior changed along the same spatial contrast; an absent educational response cannot be explained simply by an absent measured security shock.

6.2 Negotiation-period postsecondary outcomes

Figure 3 reports the common-sample event studies around 2012. The panels deliberately separate SNIES entry, short-cycle near-completion, university near-completion, and the union of the two exam-stage outcomes. Across the two designs, there is no common positive movement.

SNIES entry does not rise significantly in either comparison. For entry within one, two, and six years, every pooled confidence interval includes zero. The linked enrollment data therefore provide no statistically significant evidence that negotiations accelerated or expanded first entry into higher education.



Municipality and cohort fixed effects with baseline Saber 11 mathematics and reading controls. The shaded bands are 95 percent confidence intervals from municipality-clustered standard errors.

Figure 3: Postsecondary outcomes after negotiations begin

Notes: Coefficients are treatment-by-cohort interactions relative to 2011. Models include municipality and cohort fixed effects and baseline Saber 11 mathematics and reading controls. Shaded bands are 95 percent confidence intervals from municipality-clustered standard errors. The public SNIES crosswalk begins in 2010, so the figure contains only one pre-treatment coefficient.

There is likewise no statistically significant positive effect on exam-stage progression. In the PDET comparison, short-cycle progression falls significantly by 1.49 percentage points, while professional university and overall tertiary progression are statistically unchanged. In the FARC versus ELN comparison, the short-cycle, professional university, and overall effects are not statistically significant. Neither design therefore provides evidence of greater negotiation-period progression toward a postsecondary credential.

The event paths and Table 1 also clarify what can and cannot be inferred. Most 2012 pre-trend tests do not reject, but they compare only the 2010 coefficient with the 2011 reference year. Moreover, the 2009 to 2018 university-only specification in Appendix Table 3 rejects the FARC-ELN pre-trend because the 2009 gap differs from the 2011 gap. Failure to reject a coefficient of zero is not proof that the causal effect is exactly zero. The evidence supports the narrower conclusion that there is no statistically credible, large, and common positive response across the two designs.

6.3 Post-agreement postsecondary outcomes

Figure 4 uses 2015 as the omitted cohort and provides a longer pre-treatment period. The most notable result is the professional university path in PDET territories. Pre-treatment coefficients fluctuate close to zero, and the post coefficients rise from 0.81 percentage points in 2016 to 1.15 in 2017 and 1.42 in 2018. The pooled coefficient is 0.961 with a standard error of 0.426, which is statistically significant at the 5 percent level ($p = 0.024$). The joint pre-trend test has a p value of 0.568 and therefore does not reject equality of the pre-treatment coefficients.

This result is interesting but requires careful interpretation. PDET status was assigned in 2017 to places selected for conflict exposure, poverty, rurality, state weakness, and illicit economies. The relative increase begins with the 2016 cohort, before formal PDET designation. It therefore cannot be interpreted as the causal effect of the PDET program. The coefficient may capture agreement-era expectations, differential recovery in places subsequently prioritized, early implementation around the agreement, or changing selection. The corresponding FARC versus ELN university coefficient is 0.441 with a standard error of 0.579 and is not statistically significant; its pre-trend test rejects with a p value of 0.039. The design most directly tied to FARC demobilization therefore does not validate the PDET pattern as a general security effect.

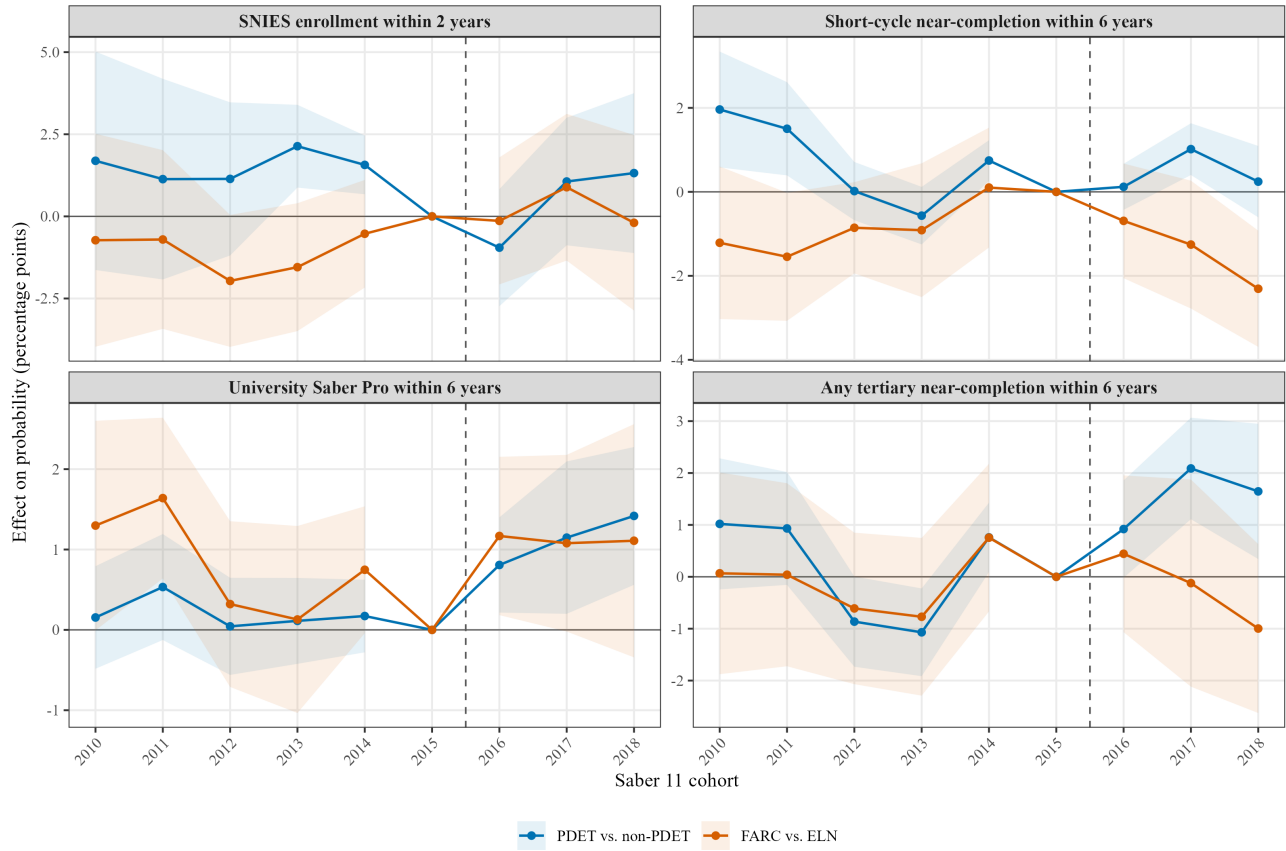
The other post-agreement outcomes reinforce that caution. PDET entry within one and two years falls, but both outcomes have differential pre-trends. Entry within six years is statistically unchanged. Overall tertiary near-completion rises, but its pre-trend test rejects strongly; short-cycle progression is also statistically unchanged. In FARC relative to ELN areas, none of these pooled outcomes changes significantly. In plain terms, the only positive post-agreement result accompanied by stable pre-treatment coefficients is professional university near-completion among PDET-origin students. Because initial enrollment and short-cycle progression do not show the same pattern, the evidence does not support a system-wide expansion of postsecondary access.

6.4 Secondary exit participation and achievement

Secondary-school outcomes help locate the pipeline response. PDET municipalities have 0.69 additional Saber 11 test takers per thousand residents relative to non-PDET municipalities after the agreement and 3.34 additional represented schools per 100,000 residents, although the school estimate is only marginally precise. Mean mathematics falls by 0.055 standard deviations and mean verbal performance

Peace agreement in 2016

Event-study estimates across the postsecondary pipeline



Municipality and cohort fixed effects with baseline Saber 11 mathematics and reading controls. The shaded bands are 95 percent confidence intervals from municipality-clustered standard errors.

Figure 4: Postsecondary outcomes around the final peace agreement

Notes: Coefficients are treatment-by-cohort interactions relative to 2015. Models include municipality and cohort fixed effects and baseline Saber 11 mathematics and reading controls. Shaded bands are 95 percent confidence intervals from municipality-clustered standard errors. The 2016 Saber 11 cohort makes its postsecondary transition as the final agreement enters into force.

Table 1: Enrollment and exam-stage progression across the postsecondary pipeline

Policy window	Outcome	PDET vs. non-PDET	Pre-trend p	FARC vs. ELN	Pre-trend p
Negotiations, 2012	SNIES enrollment within 2 years	-0.525 (1.096)	0.331	0.230 (1.538)	0.980
Negotiations, 2012	Short-cycle near-completion within 6 years	-1.495 (0.543)	0.100	0.522 (0.733)	0.591
Negotiations, 2012	University Saber Pro within 6 years	0.200 (0.269)	0.138	-0.809 (0.414)	0.502
Negotiations, 2012	Any tertiary near-completion within 6 years	-0.442 (0.422)	0.776	-0.240 (0.820)	0.964
Agreement, 2016	SNIES enrollment within 2 years	-0.785 (0.668)	< 0.001	1.095 (1.427)	0.113
Agreement, 2016	Short-cycle near-completion within 6 years	-0.123 (0.357)	< 0.001	-0.698 (0.533)	0.254
Agreement, 2016	University Saber Pro within 6 years	0.961 (0.426)	0.568	0.441 (0.579)	0.039
Agreement, 2016	Any tertiary near-completion within 6 years	1.438 (0.515)	< 0.001	-0.144 (0.930)	0.372

Notes: Each coefficient is a pooled difference-in-differences estimate in percentage points, with a municipality-clustered standard error in parentheses. All models include municipality and Saber 11 cohort fixed effects and baseline mathematics and reading controls. The 2012 models omit 2011 and have only the 2010 pre-treatment coefficient because the public Saber 11-SNIES crosswalk begins in 2010. The 2016 models omit 2015 and test the 2010-2014 coefficients jointly. Saber Pro and Saber TyT indicate near-completion, not formal graduation.

by 0.031 standard deviations.

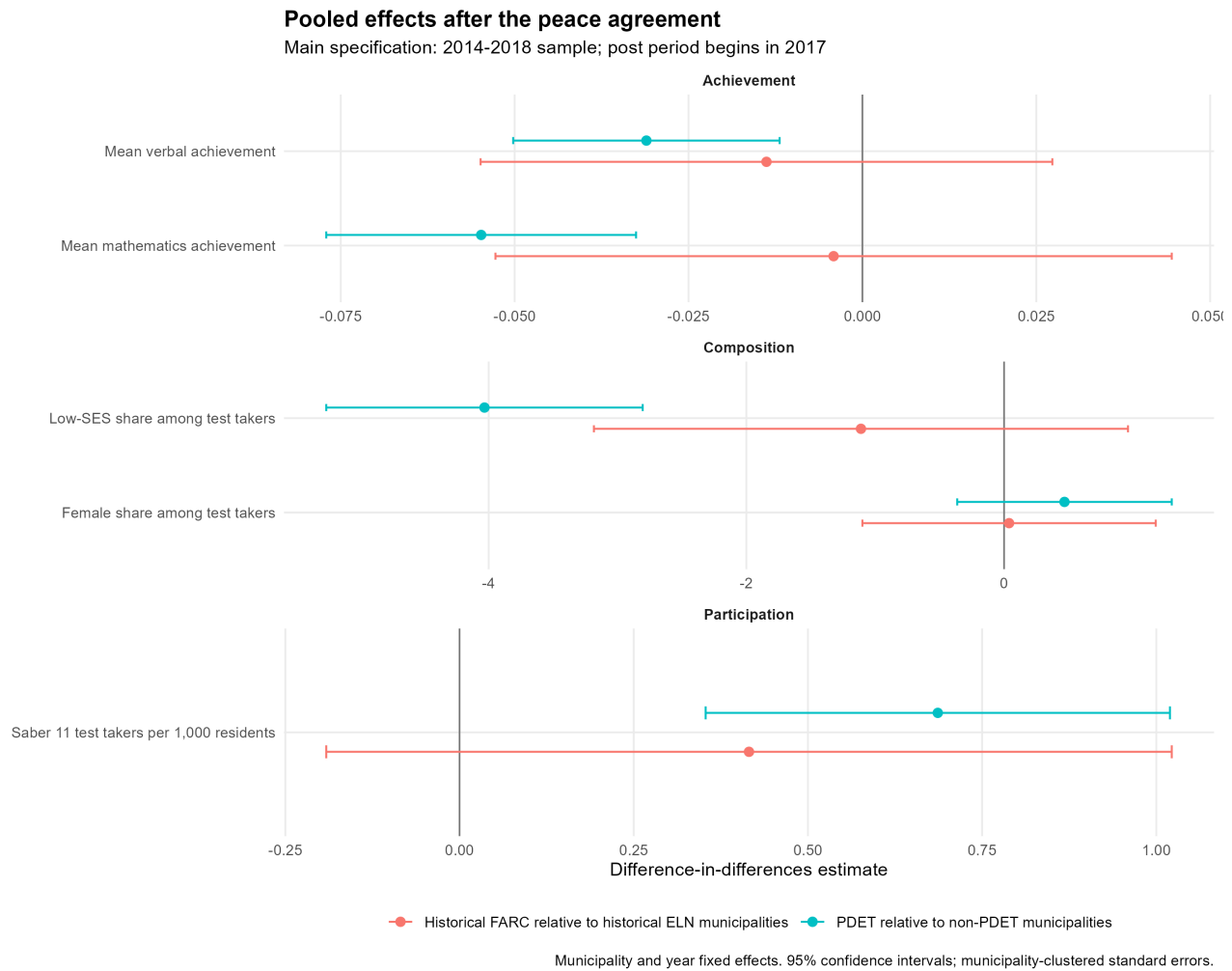


Figure 5: Pooled post-agreement effects on Saber 11 participation and achievement

Notes: Points report treatment-by-post-agreement coefficients from municipality-level models with municipality and year fixed effects. The specification uses the redesigned Saber 11 era and treats 2017 as the first full post-agreement year. Scores are standardized within cohort. Standard errors are clustered by municipality.

This pattern is consistent with changing composition, not proof that school quality deteriorated. If peace or territorial policy enables more disadvantaged or rural students to remain in school long enough to take Saber 11, average scores among examinees may fall even when no continuing student learns less. The exam records do not observe the population of youth who never reach Saber 11, so they cannot fully separate individual learning from selection into testing.

The FARC versus ELN estimates remain close to zero. Test taking rises by 0.42 per thousand residents with a standard error of 0.31. Mathematics changes by -0.004 standard deviations and verbal performance by -0.014 , both imprecise. The stronger post-agreement PDET university result therefore coexists with no clear achievement gain at secondary exit.

6.5 Timing, value added, and conditional trajectories

The additional Saber Pro analyses provide mechanism checks rather than extensive-margin evidence; [Appendix Section E](#) reports the timing and financing results. Among students reaching university-level Saber Pro within six years, elapsed time is the number of calendar years from Saber 11 to the first observed Saber Pro appearance. PDET-origin students take about 0.055 additional years after 2012, or 0.65 months. The pre-trend test has a p value of 0.069, indicating some evidence of differential pre-treatment movement. FARC versus ELN elapsed time is effectively unchanged. Excess elapsed time relative to the semester of study reported by the student, a proxy that mixes delayed entry, interruption, part-time study, transfer, program change, and reporting error, does not move consistently.

Credit financing changes by 0.35 percentage points in the PDET comparison and 0.95 points in the FARC versus ELN comparison. The delay-or-interruption proxy changes by -0.48 and 1.15 points, respectively. None is precisely estimated. Because financing and semester are observed only among eventual Saber Pro examinees, these coefficients describe the composition of selected near-completers.

Residual value added between Saber 11 and Saber Pro is similarly fragile. Pooled PDET associations are 0.049 standard deviations for quantitative performance and 0.101 for reading. The reading event study rejects stable pre-trends ($p = 0.007$), while the quantitative pre-trend test has a p value of 0.062 and also raises concern about differential pre-treatment movement. Both outcomes condition on reaching Saber Pro. FARC versus ELN estimates are -0.021 and 0.003 standard deviations. The data do not support a broad learning dividend near the end of university.

Migration, institution type, and field results also fail to reveal a common mechanism. Among FARC-origin near-completers, the public-university share is lower than among ELN-origin near-completers after 2012, but that conditional association can reflect selection and geographic supply. Neither comparison shows a robust movement into STEM. Education programs decline among later PDET near-completers. The appendix reports these paths because they characterize the selected group, not because they identify effects for the full eligible Saber 11 cohort.

7 Interpretation

The combined evidence supports two interpretive conclusions. First, SNIES shows directly that the absence of broad progression gains is not masking a clear entry boom. Enrollment within one, two, or six years does not increase consistently in either spatial design. The negotiation-period security dividend therefore did not transmit through a simple access channel visible in the linked records.

Second, the post-agreement PDET university result prevents an equally simple conclusion that nothing changed. Professional near-completion rises among students from places later prioritized for territorial development, with a stable pre-period path in the 2016 specification. Because the increase begins with the 2016 cohort and PDET was formally designated in 2017, it is not evidence that the PDET program itself caused the change. The absence of the same pattern in historical FARC relative to ELN municipalities also makes it difficult to attribute the increase to security alone. At most, the result identifies a promising pattern in places subsequently prioritized and motivates, rather than proves, a complementarity interpretation.

This distinction fits a state-capacity interpretation. [Fajardo-Steinhäuser \(2024\)](#) documents a large security improvement in historical FARC areas without a comparable expansion in broad economic activity and relates the gap to weak state entry. This paper finds the same asymmetry at the educational margin. Lower violence changes the feasible set, but families still require preparation, transport, finance,

information, and institutions capable of supplying places and supporting persistence. The positive but temporary firm-entry response documented by [Bernal et al. \(2024\)](#) likewise suggests that expectations and complementary programs mediate the response to de-escalation.

The evidence also reconciles this paper with studies finding school-level peace dividends. Primary or secondary enrollment, dropout, and achievement are earlier margins than higher-education entry or near-completion. They concern different populations and require different inputs ([Namen et al., 2021](#); [Montaño-Bardales and Palacios, 2025](#); [Avendaño Santiago, 2021](#)). Peace can improve school attendance without generating the resources required to enter and persist through a tertiary program. Conversely, a PDET university response can arise through migration, financing, or targeted institutional expansion even when average Saber 11 scores do not rise.

Uncertainty is not the paper’s principal mechanism. The SNIES data provide direct enrollment timing, and those outcomes do not show the temporary postponement pattern that the uncertainty hypothesis predicts. Semester, financing, and elapsed-time proxies also fail to move coherently. The paper therefore treats uncertainty as a possible behavioral channel that is not supported by the available estimates.

The most defensible conclusion is about transmission. The peace process produced a clear reduction in insurgent violence. It did not produce a general and design-robust improvement across the postsecondary pipeline. A suggestive professional university gain appears among students from municipalities later selected for PDET, but it begins before formal designation and cannot be attributed to PDET implementation. This pattern is consistent with, but does not establish, complementarity between security and educational or state capacity.

8 Threats to Identification and Scope

The first limitation is parallel trends. The common 2012 pipeline analysis has only the 2010 lead because the public SNIES linkage starts with the 2010 Saber 11 cohort. This is too little pre-treatment variation to validate the design strongly. The 2016 analysis has a longer pre-period, but several PDET and FARC versus ELN outcomes still reject. Some individual estimates differ statistically from zero, including the PDET short-cycle decline during negotiations and the PDET professional-university increase after the agreement, but neither forms a consistent positive pattern across designs and outcomes. For the remaining outcomes, failure to reject zero does not prove that the causal effect is exactly zero. The defensible conclusion is an absence of credible, design-robust evidence of broad positive effects.

Second, PDET status is a post-agreement policy classification. It is a marker for territories prioritized by the peace architecture, not a treatment randomly assigned in 2012 or 2016. PDET estimates can reflect poverty, rurality, coca exposure, conflict, institutional weakness, later investment, or differential recovery. Even an event path with stable pre-trends does not isolate which component generated the change.

Third, the FARC versus ELN design has only 41 control municipalities. Historical presence can be measured with error, and the ELN is not a sealed untreated group. ELN territories experienced common national reforms, spillovers, changing armed competition, and their own political negotiations. The design estimates a relative effect between two conflict exposures rather than peace relative to an unchanging war counterfactual.

Fourth, SNIES linkage begins in 2010. Students from the 2009 Saber 11 cohort remain usable in Saber Pro analysis but cannot be assigned enrollment outcomes. Treating them as zero would confuse dataset noncoverage with nonenrollment. The paper therefore uses 2010 to 2018 whenever entry

and progression are compared on a common population and 2009 to 2018 only for explicitly labeled university robustness checks.

Fifth, Saber Pro and Saber TyT are not formal graduation records. They identify students who reach the examination stage near the end of a program. Some examinees may not graduate. The harmonization removes the 2016 program-sector break, but program-level classification in historical files can still contain error. The short-cycle and university outcomes should be interpreted as advanced progression rather than credentials earned.

Sixth, municipality is fixed at Saber 11. The results measure the opportunity of youth originating in a municipality and not local supply alone. Students may migrate, and the linked SNIES crosswalk records the first institution rather than a complete history of all transfers and interruptions. A future extension using full SNIES histories or the *Sistema para la Prevención de la Deserción de la Educación Superior* (SPADIES), the Ministry of Education's longitudinal system for monitoring higher-education enrollment and dropout, could distinguish local entry, migration, persistence, and formal completion.

Seventh, the analysis conditions on taking Saber 11 in the regular second-semester session. This restriction is valuable for identifying recent school leavers, but peace may also affect whether youth finish secondary school or take the exit examination. The municipality-level participation results partially address that margin; a full welfare analysis would start from population cohorts rather than test takers.

Finally, the conflict panel measures reported events rather than latent territorial control. Reporting can change with police presence, road access, or institutional capacity. The core index reduces reliance on any one category and has stable FARC versus ELN pre-trends, but an administrative zero does not prove an absence of insecurity.

9 Conclusion

Colombia's peace process generated a large security dividend and a much less general educational dividend. Historical FARC municipalities experienced a sharp and persistent decline in core conflict relative to historical ELN municipalities. The linked SNIES data show no corresponding increase in higher-education entry during negotiations, and the combined Saber TyT and Saber Pro records show no increase in overall exam-stage progression. Secondary achievement, value added, timing, financing, migration, and field choice also fail to improve consistently across designs.

The main exception is professional university progression in municipalities later selected for PDET. Students from these municipalities become about one percentage point more likely to reach university-level Saber Pro within six years, and the pre-agreement event path is stable in the 2016 specification. The same gain does not appear credibly in historical FARC relative to ELN municipalities. Because the increase begins before formal PDET designation, the coefficient is best read as a suggestive differential response in places subsequently prioritized, not as the effect of the PDET program or of security alone.

The paper's contribution is therefore not a declaration that peace had no human-capital consequences. It is a map of where those consequences are and are not visible. Security improved clearly. Entry and broad tertiary progression did not respond generally. A narrower professional university gain appeared only in prioritized territories. This pattern implies that peace removes a major constraint but does not automatically create the educational capacity required for sustained human-capital accumulation.

A Baseline Characteristics and the University-Only Sample

Table 2 reports baseline characteristics for the 2009 to 2011 university-only sample. PDET places are substantially poorer and have lower initial achievement and university near-completion than non-PDET places. FARC and ELN municipalities are closer in achievement levels but still differ in socioeconomic composition. These level differences motivate fixed effects and reinforce the need to evaluate trends.

Table 2: Baseline characteristics, 2009 to 2011

Design	Characteristic	Treated	Comparison	Std. difference
PDET	Female student	56.0%	54.8%	0.02
PDET	Household stratum 1 or 2	91.6%	73.7%	0.49
PDET	Saber 11 mathematics, cohort SD	-0.27	0.03	-0.32
PDET	Saber 11 verbal achievement, cohort SD	-0.26	0.03	-0.30
PDET	University-level Saber Pro within six years	10.8%	15.9%	-0.15
PDET	Years to university-level Saber Pro	5.29	5.21	0.11
FARC-ELN	Female student	55.2%	56.5%	-0.03
FARC-ELN	Household stratum 1 or 2	91.8%	86.0%	0.19
FARC-ELN	Saber 11 mathematics, cohort SD	-0.17	-0.14	-0.03
FARC-ELN	Saber 11 verbal achievement, cohort SD	-0.16	-0.16	0.00
FARC-ELN	University-level Saber Pro within six years	10.9%	9.7%	0.04
FARC-ELN	Years to university-level Saber Pro	5.26	5.30	-0.06

Notes: Treated denotes PDET or historical FARC municipalities. The comparison denotes non-PDET or historical ELN municipalities. Years are conditional on reaching university-level Saber Pro within six years. Standardized differences divide the difference in means by the pooled individual-level standard deviation.

Table 3 reports the complementary 2009 to 2018 university-only analysis without the SNIES common-sample restriction. It shows that including 2009 and using the years-four-through-six robustness definition does not produce a credible negotiation-period university gain. These coefficients should not be mixed mechanically with Table 1, which uses 2010 to 2018, baseline score controls, and a common population for SNIES, TyT, and Pro outcomes.

B Additional Descriptive Evidence

Figure 6 displays raw standardized achievement by cohort. The persistent PDET gap makes clear why levels are not informative. FARC and ELN trends overlap more closely, although the smaller ELN group produces less precise estimates.

Table 3: Selected educational estimates using a consistent university-level outcome

Outcome and comparison	Estimate	Std. error	Pre-trend <i>p</i>
<i>Panel A: PDET relative to non-PDET municipalities</i>			
University-level Saber Pro within six years, pooled post 2012, pp	0.453	0.270	0.000
University-level Saber Pro in years four to six, pooled post 2012, pp	0.519	0.264	0.000
University-level Saber Pro within six years, 2012 event, pp	−0.492	0.318	0.000
University-level Saber Pro within six years, 2018 event, pp	0.880	0.487	0.000
Years from Saber 11 to university-level Saber Pro, pooled post 2012	0.055	0.012	0.069
University quantitative value added, pooled post 2012, SD	0.049	0.015	0.062
University reading value added, pooled post 2012, SD	0.101	0.024	0.007
<i>Panel B: Historical FARC relative to historical ELN municipalities</i>			
University-level Saber Pro within six years, pooled post 2012, pp	−0.432	0.417	0.001
University-level Saber Pro in years four to six, pooled post 2012, pp	−0.409	0.415	0.000
University-level Saber Pro within six years, 2012 event, pp	−1.312	0.568	0.001
University-level Saber Pro within six years, 2018 event, pp	−0.567	0.741	0.001
Years from Saber 11 to university-level Saber Pro, pooled post 2012	−0.007	0.022	0.478
University quantitative value added, pooled post 2012, SD	−0.021	0.020	0.183
University reading value added, pooled post 2012, SD	0.003	0.024	0.096

Notes: Binary outcomes are percentage points. All eligible Saber 11 students remain in the university-outcome denominator; nonmatches, technical or technological records, and university records outside the horizon equal zero. Event coefficients omit 2011. Models include municipality and cohort fixed effects and cluster standard errors by municipality. Pre-trend tests evaluate 2009 and 2010 jointly. Conditional duration and value-added estimates are selected-completer outcomes.

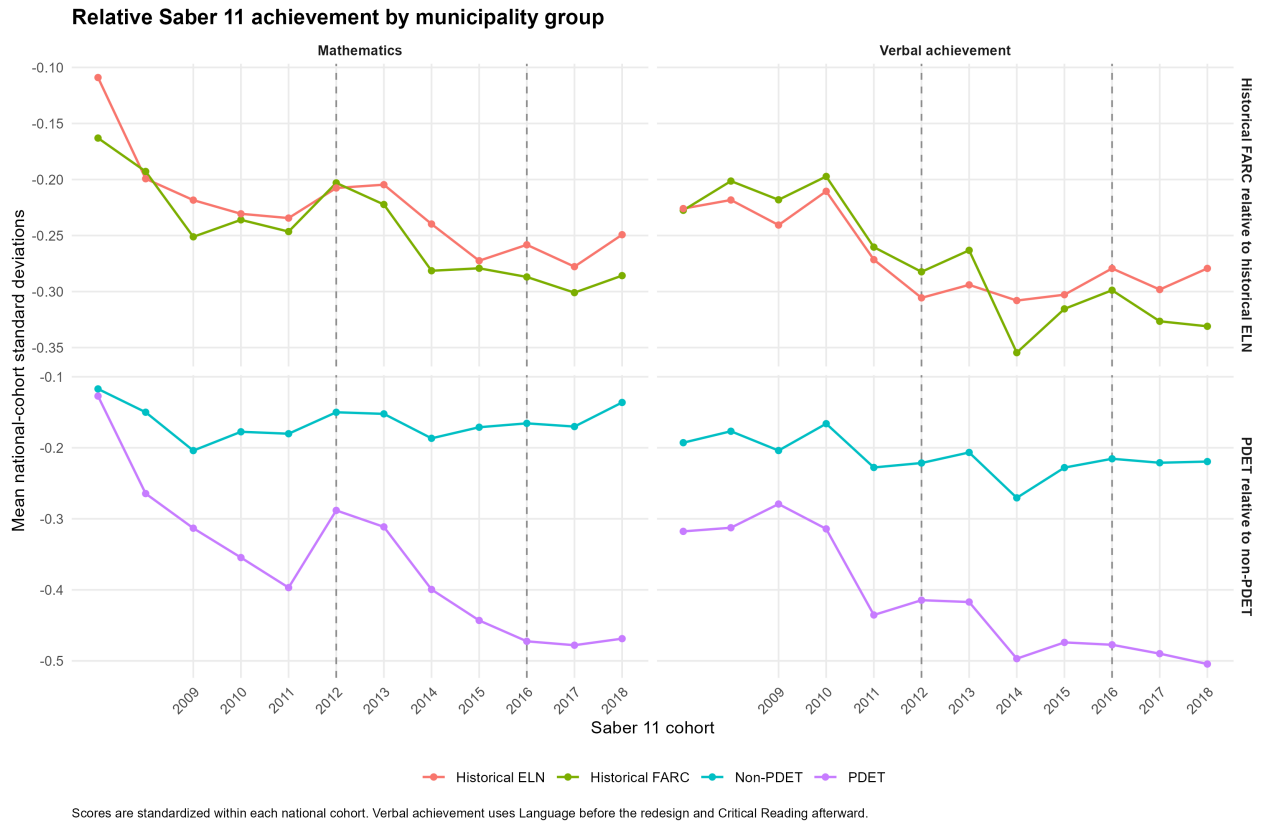
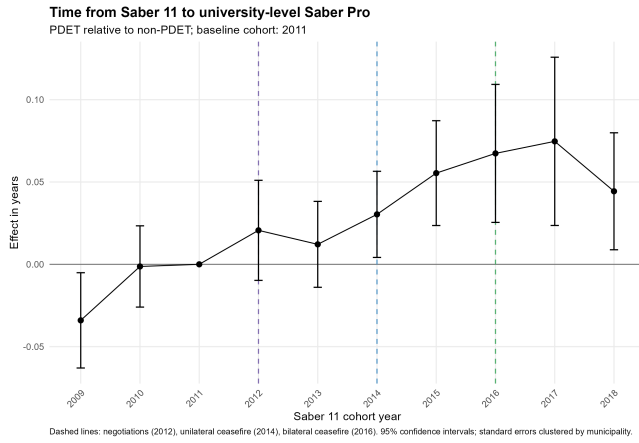


Figure 6: Raw Saber 11 achievement trends

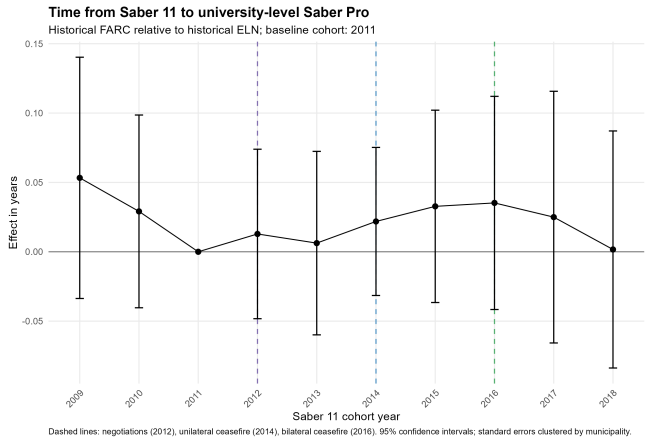
Notes: Mathematics and verbal scores are standardized within examination cohort. Lines report group means before regression adjustment.

C Conditional University Trajectories

The following figures describe students observed in university-level Saber Pro. Because conditioning on progression can induce selection, they characterize the composition of near-completers rather than causal outcomes for the full eligible Saber 11 cohort.



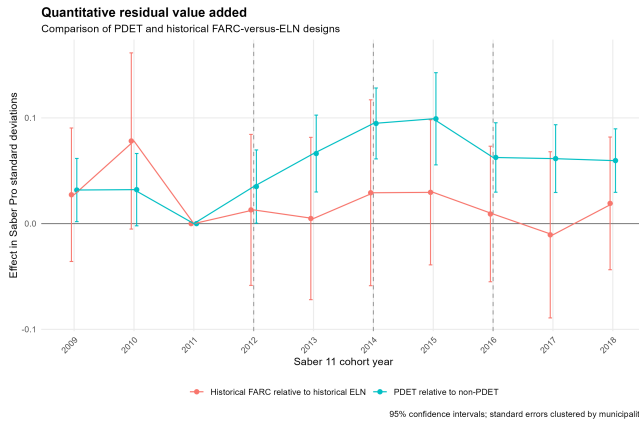
(a) PDET relative to non-PDET



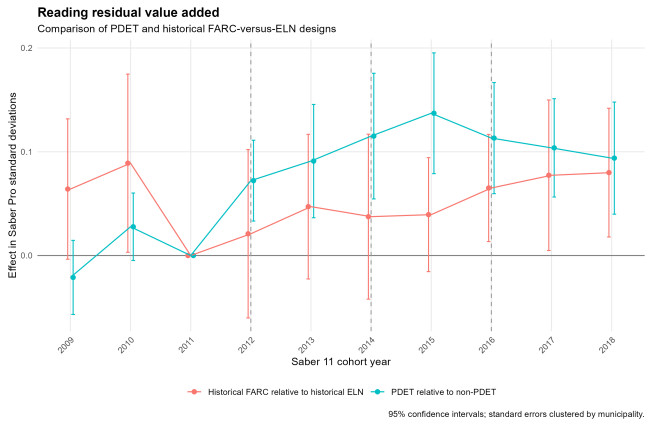
(b) Historical FARC relative to historical ELN

Figure 7: Time from Saber 11 to university-level Saber Pro

Notes: The sample conditions on appearing in university-level Saber Pro within six years. Coefficients are measured in years relative to the omitted 2011 cohort. The PDET and FARC versus ELN pre-trend p values are 0.069 and 0.479.



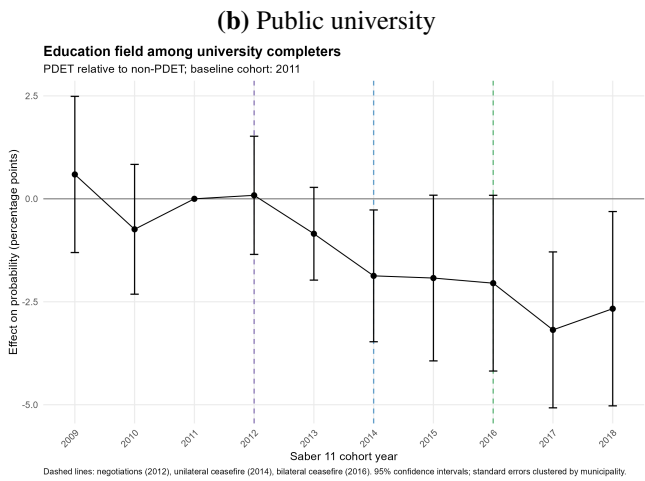
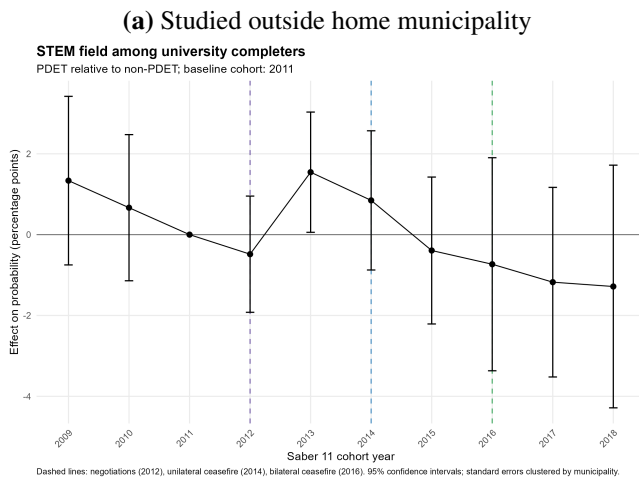
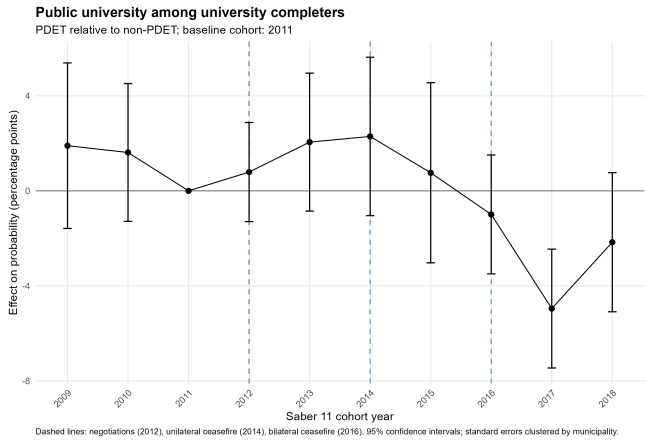
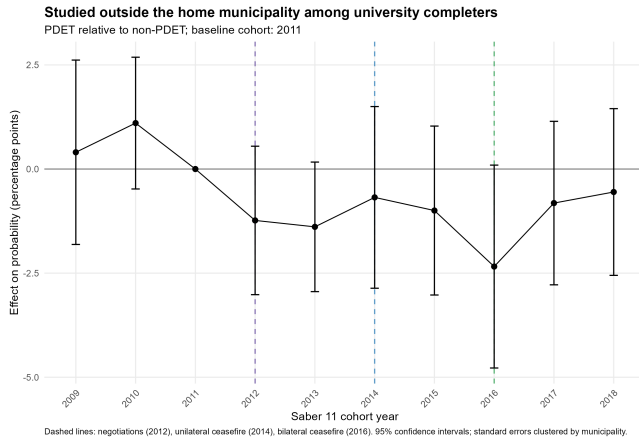
(a) Quantitative residual value added



(b) Reading residual value added

Figure 8: Value added between Saber 11 and Saber Pro

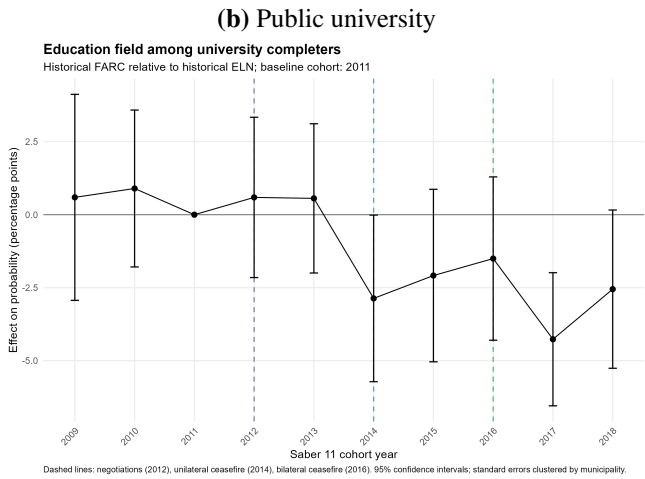
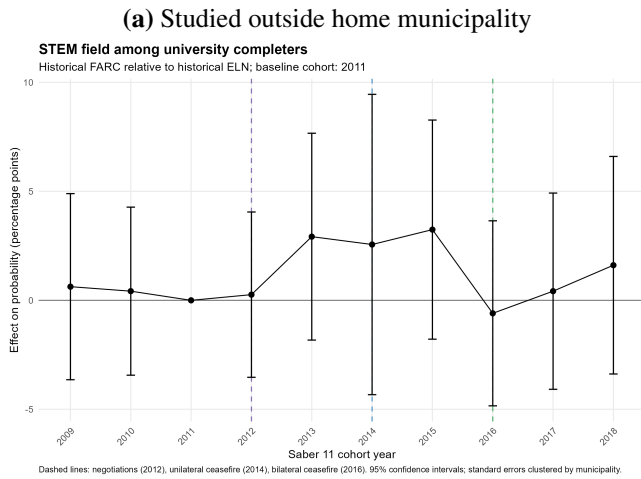
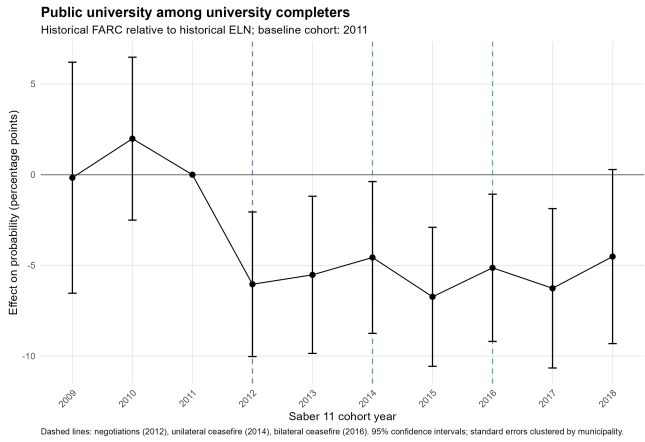
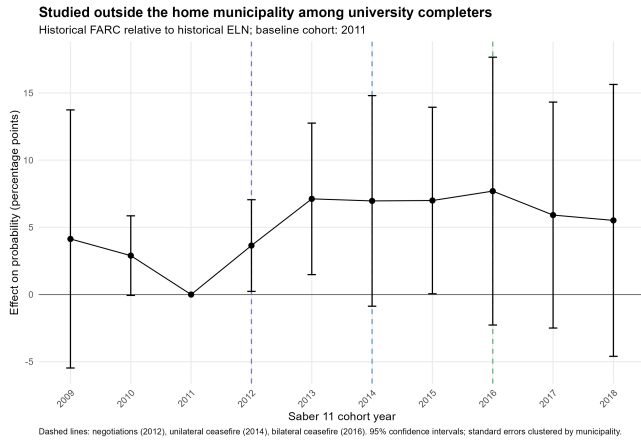
Notes: Saber Pro scores are standardized within test year and residualized on the corresponding Saber 11 score and baseline covariates. Estimates condition on reaching university-level Saber Pro within six years. The PDET pre-trend p values are 0.062 for quantitative and 0.007 for reading value added. The FARC versus ELN values are 0.183 and 0.096.



(c) STEM field

(d) Education field

Figure 9: Composition among PDET-origin university near-completers



(c) STEM field

(d) Education field

Figure 10: Composition among historical FARC and ELN university near-completers

D Conflict Outcomes and Construction

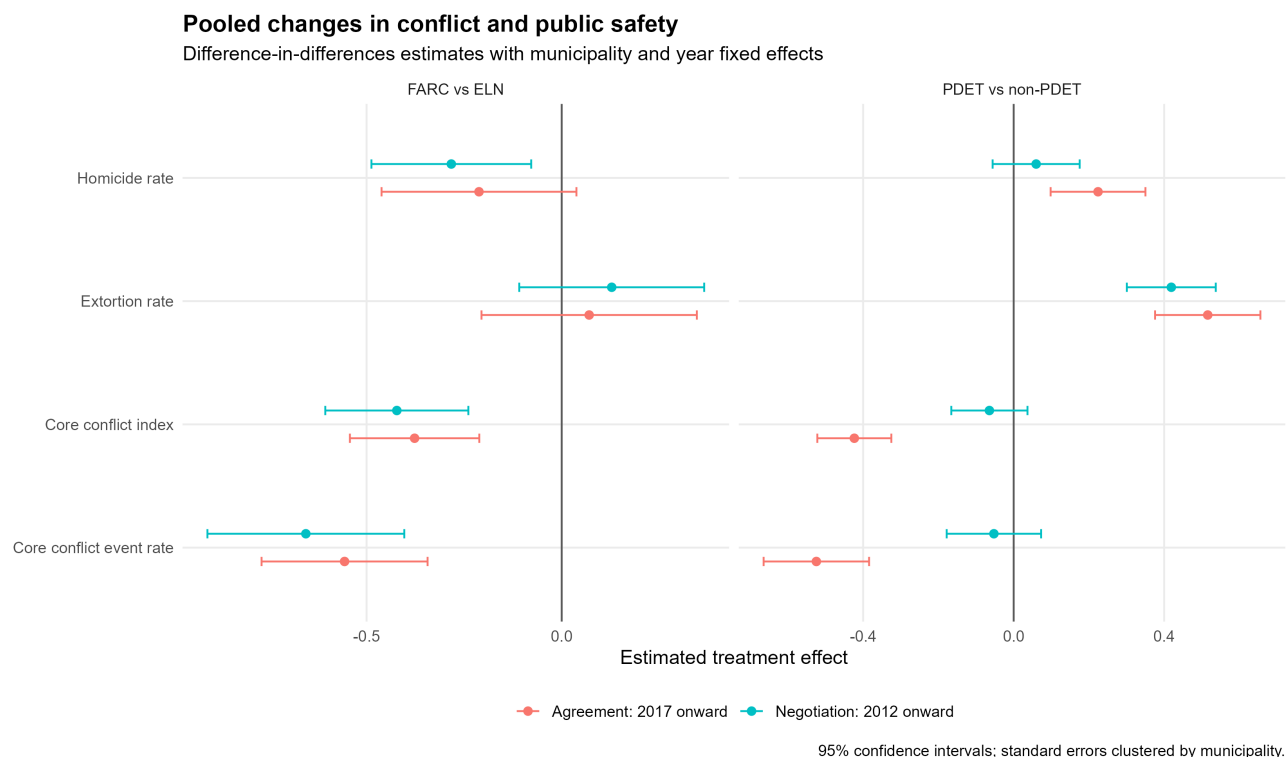


Figure 11: Pooled conflict and public-safety regressions

Notes: Points report treatment-by-policy-period coefficients from municipality-year models with municipality and year fixed effects. Equal municipality weights are used. Horizontal bars are 95 percent confidence intervals from municipality-clustered standard errors.

Table 4: Main conflict regressions

Design	Outcome	Policy period	Estimate (SE)
PDET vs non-PDET	Core conflict index	Negotiation, 2012 onward	-0.064 (0.052)
PDET vs non-PDET	Core conflict event rate	Negotiation, 2012 onward	-0.053 (0.064)
PDET vs non-PDET	Core conflict index	Agreement, 2017 onward	-0.423 (0.050)
PDET vs non-PDET	Core conflict event rate	Agreement, 2017 onward	-0.524 (0.072)
FARC vs ELN	Core conflict index	Negotiation, 2012 onward	-0.422 (0.094)
FARC vs ELN	Core conflict event rate	Negotiation, 2012 onward	-0.656 (0.129)
FARC vs ELN	Core conflict index	Agreement, 2017 onward	-0.377 (0.085)
FARC vs ELN	Core conflict event rate	Agreement, 2017 onward	-0.556 (0.109)

Notes: All models include municipality and year fixed effects. Standard errors in parentheses are clustered by municipality. Dependent variables are defined in Code 10.

Notes: Each entry reports a treatment-by-post coefficient followed by its municipality-clustered standard error in parentheses. All specifications contain municipality and year fixed effects. The core index averages standardized log population-normalized rates for the five event categories with complete 2007 to 2018 coverage.

D.1 Conflict-variable construction

D.1.1 Annual conflict outcomes

The annual outcomes are constructed from ten event-level administrative files produced by the Colombian Ministry of Defense’s National Police, covering conflict-related incidents and broader public-safety events. I aggregate the records to the municipality-year level for 2007 to 2018. Municipal population, used only to construct rates, comes from CEDE’s General Characteristics Panel. These police files measure outcomes during the analysis period and are not the source used to classify municipalities as historically FARC or ELN.

Let C_{jmt} denote the count for event family j in municipality m and year t , and let Pop_{mt} be population. For the five families with complete 2007 to 2018 coverage, I construct

$$r_{jmt} = 100,000 \frac{C_{jmt}}{Pop_{mt}}, \quad z_{jmt} = \frac{\log(1 + r_{jmt}) - \bar{\ell}_j}{s_j}, \quad ConflictIndex_{mt} = \frac{1}{5} \sum_{j=1}^5 z_{jmt}.$$

The five families are subversive actions, attacks on bridges and roads, kidnapping, terrorism, and oil-pipeline bombing. Attacks on security forces are excluded because that file begins in 2010. Homicide, extortion, and land invasion are treated as broader safety measures. Cocaine observations are not used as a main outcome because the source records enforcement quantities rather than coca cultivation or production. A zero is assigned only to municipality-years inside the observed coverage period of each source.

D.1.2 Historical FARC and ELN classification

CEDE is the Centro de Estudios sobre Desarrollo Económico, the economics research center at Universidad de los Andes. Its DataHub distributes harmonized municipal panels assembled from Colombian administrative and statistical sources. I use two CEDE datasets. The 2021 municipal Conflict and Violence Panel supplies the historical actor-attributed events used to classify municipalities, while the General Characteristics Panel supplies municipal population. The Conflict and Violence Panel reports municipality-year event counts separately attributed to the FARC and the ELN (CEDE, 2022). It is distinct from the police event files used to measure the 2007 to 2018 conflict outcomes above. The classification follows the territorial comparison in Fajardo-Steinhäuser (2024).

For each organization, I use 23 attributed event families: attacks against the civilian population, incursions into populated centers, assaults against the population, incendiary and explosive terrorist acts, offensive actions, kidnappings of politicians, members of the armed forces, and civilians, attacks against transport, illegal checkpoints, armed harassment, attacks against institutions or official entities, illegal roadblocks, assaults against private property, political attacks, ambushes, clashes, armed contacts, political homicides, members of the armed forces wounded or killed, and insurgent-group members killed. Let A_{kmt}^g be the count for event family k attributed to organization $g \in \{FARC, ELN\}$ in municipality m and year t . A group is active in a municipality-year when at least one attributed event is recorded:

$$Active_{mt}^g = \mathbb{1} \left\{ \sum_{k=1}^{23} A_{kmt}^g > 0 \right\}.$$

Missing actor-event entries in the selected CEDE variables are treated as zeros. The annual indicator therefore measures the extensive margin of recorded presence, not the number or severity of events.

The historical window runs from 1996 through 2008, thirteen years ending before the 2012 negotiations. I count the number of active years and define persistent historical presence as

$$Years_m^g = \sum_{t=1996}^{2008} Active_{mt}^g, \quad Historical_m^g = \mathbb{1}\{Years_m^g \geq 8\}.$$

The threshold of eight years corresponds to activity in at least 60 percent of the historical period, since $0.60 \times 13 = 7.8$. It is designed to capture sustained territorial presence rather than isolated incidents.

The final comparison is mutually exclusive. After excluding the largest two percent of municipalities by average 1996 to 2008 population, I define

$$FARC_ELN_m = \begin{cases} 1, & Historical_m^{FARC} = 1 \text{ and } Historical_m^{ELN} = 0, \\ 0, & Historical_m^{FARC} = 0 \text{ and } Historical_m^{ELN} = 1, \\ \text{missing,} & \text{otherwise.} \end{cases}$$

Thus, “historical FARC” does not require that the ELN was never observed; it requires that only the FARC satisfies the eight-year persistence threshold. The analogous interpretation applies to historical ELN municipalities. Municipalities in which both groups satisfy the threshold and municipalities in which neither group does are excluded. The resulting sample contains 216 historical FARC municipalities and 41 historical ELN municipalities. This fixed classification is merged to students using their municipality of residence when they take Saber 11.

E Timing, Financing, and Measurement Limits

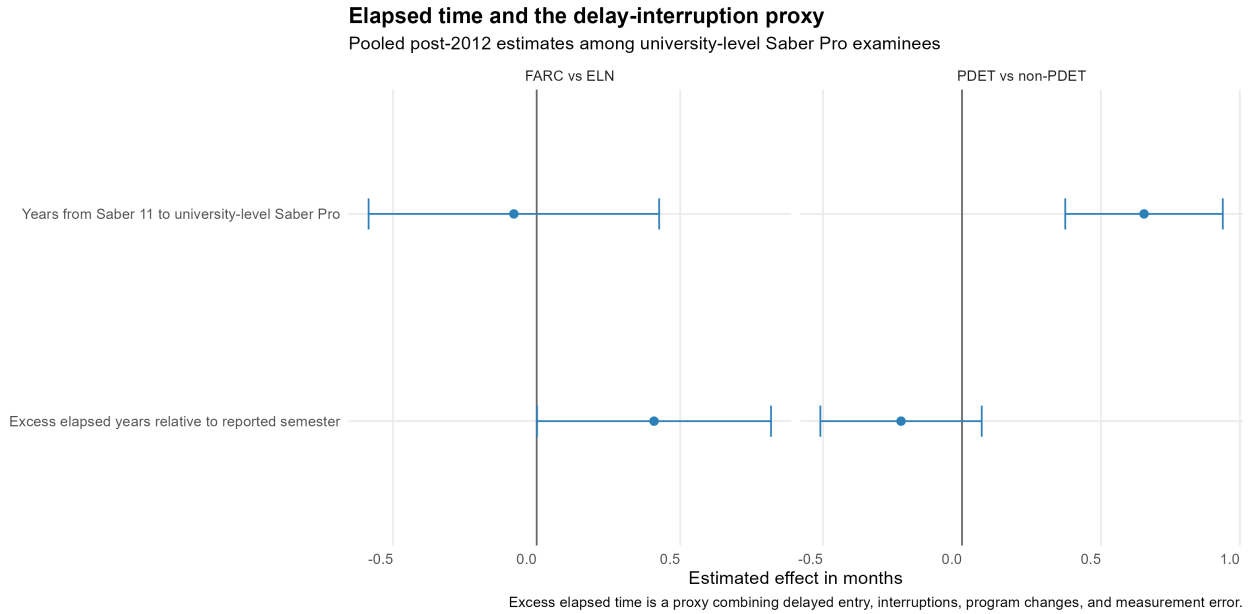


Figure 12: Elapsed time and the delay-or-interruption proxy

Notes: Models condition on reaching university-level Saber Pro within six years. Excess elapsed time subtracts one-half of the semester reported at Saber Pro from years since Saber 11. It combines delayed entry, interruptions, program changes, and measurement error.

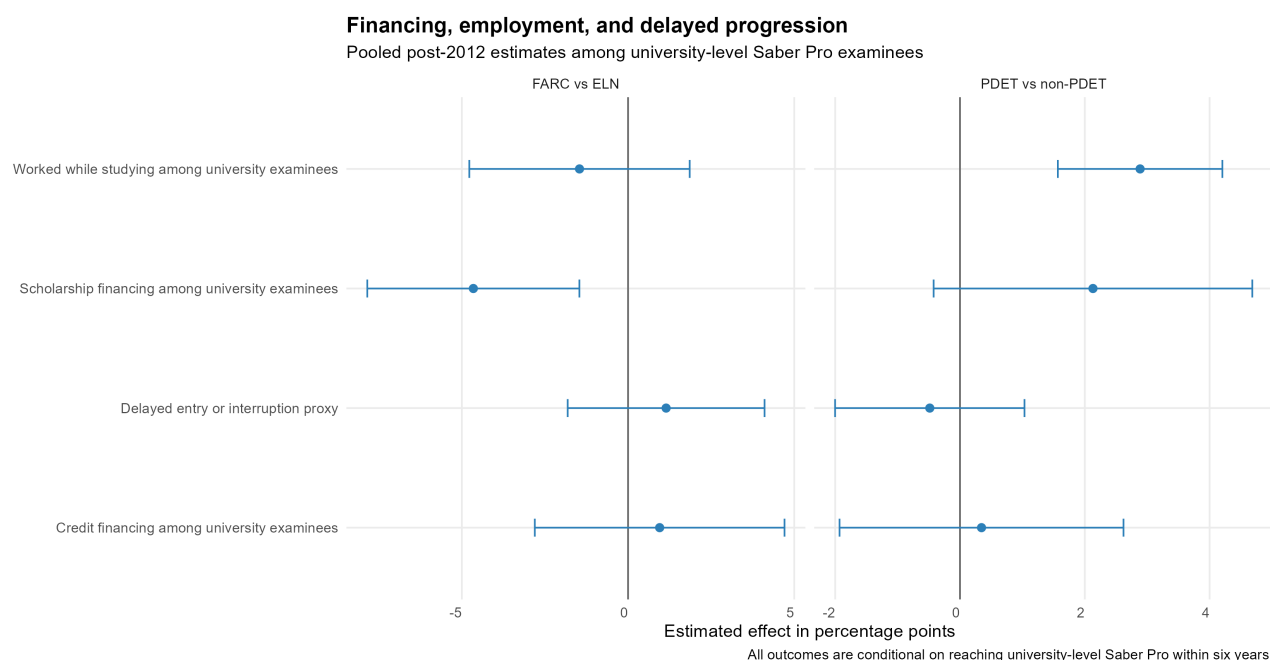


Figure 13: Financing, employment, and delayed progression among university examinees

Notes: Outcomes are measured among students reaching university-level Saber Pro within six years. Financing variables describe credit or scholarship resources. Employment records work in the week before the exam. The delay indicator equals one when excess elapsed time is at least one year.

Table 5: Pooled timing and financing regressions

Design	Outcome	Estimate	SE	Pre-trend <i>p</i>
FARC vs ELN	Credit financing among university examinees	0.951	1.917	0.539
FARC vs ELN	Delayed entry or interruption proxy	1.145	1.511	0.668
FARC vs ELN	Excess elapsed years relative to reported semester	0.034	0.017	0.718
FARC vs ELN	University-level Saber Pro within four years	-0.073	0.127	0.653
FARC vs ELN	University-level Saber Pro within five years	-0.032	0.309	0.005
FARC vs ELN	University-level Saber Pro within six years	-0.432	0.417	0.001
FARC vs ELN	University-level Saber Pro in years four to six	-0.409	0.415	0.000
PDET vs non-PDET	Credit financing among university examinees	0.345	1.161	0.860
PDET vs non-PDET	Delayed entry or interruption proxy	-0.484	0.774	0.322
PDET vs non-PDET	Excess elapsed years relative to reported semester	-0.018	0.012	0.844
PDET vs non-PDET	University-level Saber Pro within four years	-0.476	0.135	0.469
PDET vs non-PDET	University-level Saber Pro within five years	0.004	0.219	0.086
PDET vs non-PDET	University-level Saber Pro within six years	0.453	0.270	0.000
PDET vs non-PDET	University-level Saber Pro in years four to six	0.519	0.264	0.000

Notes: Binary outcomes are percentage points; excess elapsed time is years. Models include municipality and Saber 11 cohort fixed effects and baseline score controls. Conditional Saber Pro outcomes include examination-period fixed effects. These estimates use the 2009 to 2018 university-only sample.

Table 6: Observed and unobserved postsecondary margins

Margin	Status in linked data	Interpretation
SNIES first enrollment	Directly observed from the crosswalk	Entry timing after Saber 11, available for cohorts beginning in 2010
University Saber Pro within six years	Directly observed	Reaching at least 75 percent of a professional program, not formal graduation
Harmonized short-cycle exam within six years	Constructed from technical and technological Pro plus TyT	Reaching the final examination stage of a short-cycle program
Years from Saber 11 to Saber Pro	Directly observed	Total elapsed time among university near-completers
Current program semester	Self-reported in most Saber Pro years	Program position at examination, not a complete enrollment history
Excess elapsed time	Constructed proxy	Delayed entry, interruption, transfer, program change, or reporting differences
Credit and scholarship use	Reported at Saber Pro	Financing composition among selected university near-completers
Formal graduation	Not observed	Requires SNIES graduation or institutional degree records
Semester-by-semester persistence	Not observed in the released crosswalk	Requires full SNIES or SPADIES histories

References

- Avendaño Santiago, Paula Andrea. 2021. “Desescalamiento del conflicto armado y rendimiento escolar: evidencia de la recomposición estudiantil como mecanismo potencial, 2008 a 2019.” Master’s thesis, Universidad del Rosario. https://doi.org/10.48713/10336_32271.
- Becker, Gary S. 1964. *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*. Chicago: University of Chicago Press.
- Bernal, Carolina, Mounu Prem, Juan F. Vargas, and Mónica Ortiz. 2024. “Peaceful Entry: Entrepreneurship Dynamics during Colombia’s Peace Agreement.” *Journal of Development Economics* 166: 103119. <https://doi.org/10.1016/j.jdeveco.2023.103119>.
- Blattman, Christopher, and Edward Miguel. 2010. “Civil War.” *Journal of Economic Literature* 48(1): 3-57. <https://doi.org/10.1257/jel.48.1.3>.
- Government of Colombia. 2017. “Decreto Ley 893 de 2017: Por el cual se crean los Programas de Desarrollo con Enfoque Territorial, PDET.” Departamento Administrativo de la Función Pública. <https://www.funcionpublica.gov.co/eva/gestornormativo/norma.php?i=81856>.
- Fajardo-Steinhäuser, Miguel. 2024. “Peace Dividends: The Economic Effects of Colombia’s Peace Agreement.” arXiv working paper 2301.01843, revised December 2024. <https://arxiv.org/abs/2301.01843>.
- Gómez Soler, Silvia C. 2016. “Educational Achievement at Schools: Assessing the Effect of the Civil Conflict Using a Pseudo-panel of Schools.” *International Journal of Educational Development* 49: 91-106. <https://doi.org/10.1016/j.ijedudev.2016.02.004>.
- Gómez Soler, Silvia C. 2017. “Civil Conflict and Educational Achievement: The Case of the Colombian Secondary School Exit Examination.” *Colombia Internacional* 92: 73-103. <https://doi.org/10.7440/colombiaint92.2017.03>.
- Instituto Colombiano para la Evaluación de la Educación. 2026. “Acerca del examen Saber Pro.” Accessed August 10, 2026. <https://www.icfes.gov.co/evaluaciones-icfes/acerca-del-examen-saber-pro/>.
- Justino, Patricia. 2012. “War and Poverty.” In *The Oxford Handbook of the Economics of Peace and Conflict*, edited by Michelle R. Garfinkel and Stergios Skaperdas, 676-699. Oxford: Oxford University Press.
- Montaño-Bardales, Juan Fernando, and Paola Palacios. 2025. “Educational Outcomes in Times of Conflict and Post-conflict.” *International Journal of Educational Development* 113: 103193. <https://doi.org/10.1016/j.ijedudev.2024.103193>.
- Namen, Olga, Mounu Prem, and Juan F. Vargas. 2021. “The Human Capital Peace Dividend.” Working paper, SSRN 3424863. <https://doi.org/10.2139/ssrn.3424863>.
- Prem, Mounu, Santiago Saavedra, and Juan F. Vargas. 2020. “End-of-conflict Deforestation: Evidence from Colombia’s Peace Agreement.” *World Development* 129: 104852. <https://doi.org/10.1016/j.worlddev.2019.104852>.
- Rodríguez, Catherine, and Fabio Sánchez. 2012. “Armed Conflict Exposure, Human Capital Investments, and Child Labor: Evidence from Colombia.” *Defence and Peace Economics* 23(2): 161-184. <https://doi.org/10.1080/10242694.2011.597239>.
- United Nations Security Council. 2016. “Letter Dated 14 December 2016 from the Secretary-General Addressed to the President of the Security Council.” Document S/2016/1063. <https://digitallibrary.un.org/record/852151>.
- United Nations Security Council. 2017. *Final Agreement for Ending the Conflict and Building a Stable and Lasting Peace*. Document S/2017/272. <https://digitallibrary.un.org/record/1298667>.
- Athreya, Kartik, and Janice Eberly. 2021. “Risk, the College Premium, and Aggregate Human Capital Investment.” *American Economic Journal: Macroeconomics* 13(2): 168-213. <https://doi.org/10.1257/>

mac.20160396.

- Carranza, Juan Esteban, and María Marta Ferreyra. 2019. "Increasing Higher Education Access: Supply, Sorting, and Outcomes in Colombia." *Journal of Human Capital* 13(1): 95-136. <https://doi.org/10.1086/701435>.
- Chamarbagwala, Rubiana, and Hilcías E. Morán. 2011. "The Human Capital Consequences of Civil War: Evidence from Guatemala." *Journal of Development Economics* 94(1): 41-61. <https://doi.org/10.1016/j.jdeveco.2010.01.005>.
- Justino, Patricia. 2016. "Supply and Demand Restrictions to Education in Conflict-Affected Countries: New Research and Future Agendas." *International Journal of Educational Development* 47: 76-85. <https://doi.org/10.1016/j.ijedudev.2016.01.002>.
- Kazianga, Harounan. 2012. "Income Risk and Household Schooling Decisions in Burkina Faso." *World Development* 40(8): 1647-1662. <https://doi.org/10.1016/j.worlddev.2012.04.017>.
- León, Gianmarco. 2012. "Civil Conflict and Human Capital Accumulation: The Long-Term Effects of Political Violence in Peru." *Journal of Human Resources* 47(4): 991-1022.
- Londoño-Vélez, Juliana, Catherine Rodríguez, and Fabio Sánchez. 2020. "Upstream and Downstream Impacts of College Merit-Based Financial Aid for Low-Income Students: Ser Pilo Paga in Colombia." *American Economic Journal: Economic Policy* 12(2): 193-227. <https://doi.org/10.1257/pol.20180131>.
- Melguizo, Tatiana, Fabio Sánchez, and Tatiana Velasco. 2016. "Credit for Low-Income Students and Access to and Academic Performance in Higher Education in Colombia: A Regression Discontinuity Approach." *World Development* 80: 61-77. <https://doi.org/10.1016/j.worlddev.2015.11.018>.
- Shemyakina, Olga. 2011. "The Effect of Armed Conflict on Accumulation of Schooling: Results from Tajikistan." *Journal of Development Economics* 95(2): 186-200. <https://doi.org/10.1016/j.jdeveco.2010.05.002>.
- Stange, Kevin M. 2012. "An Empirical Investigation of the Option Value of College Enrollment." *American Economic Journal: Applied Economics* 4(1): 49-84. <https://doi.org/10.1257/app.4.1.49>.
- Centro de Estudios sobre Desarrollo Económico. 2022. "Panel Conflicto y Violencia." DataHub, Universidad de los Andes, Version 2. <https://doi.org/10.57924/BN57KJ>.
- Instituto Colombiano para la Evaluación de la Educación. 2014. "Acuerdo 0023 de 2014: Estructura y organización del examen de Estado Saber 11." https://normograma.icfes.gov.co/compilacion/docs/acuerdo_icfes_0023_2014.htm.
- Instituto Colombiano para la Evaluación de la Educación. 2016. "Resolución 0455 de 2016: Calendario de los exámenes Saber Pro y Saber TyT." https://normograma.icfes.gov.co/compilacion/docs/resolucion_icfes_0455_2016.htm.